DMFC for Windows user's manual

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Index

DMFC for Windows is 1

System requirements 1

Communication Hardware 2

Software functions 2

Parameter Setting 2

Communication Parameters Tap **2**

Data save function 3

Re-search function 4

Parameter configuration tap 5

Main windows' Multi View tap 8

Main windows' History View tap 9

Appendix 11

Appendix A. Plot toolbar functions 11

Appendix B. Communication results and error 12

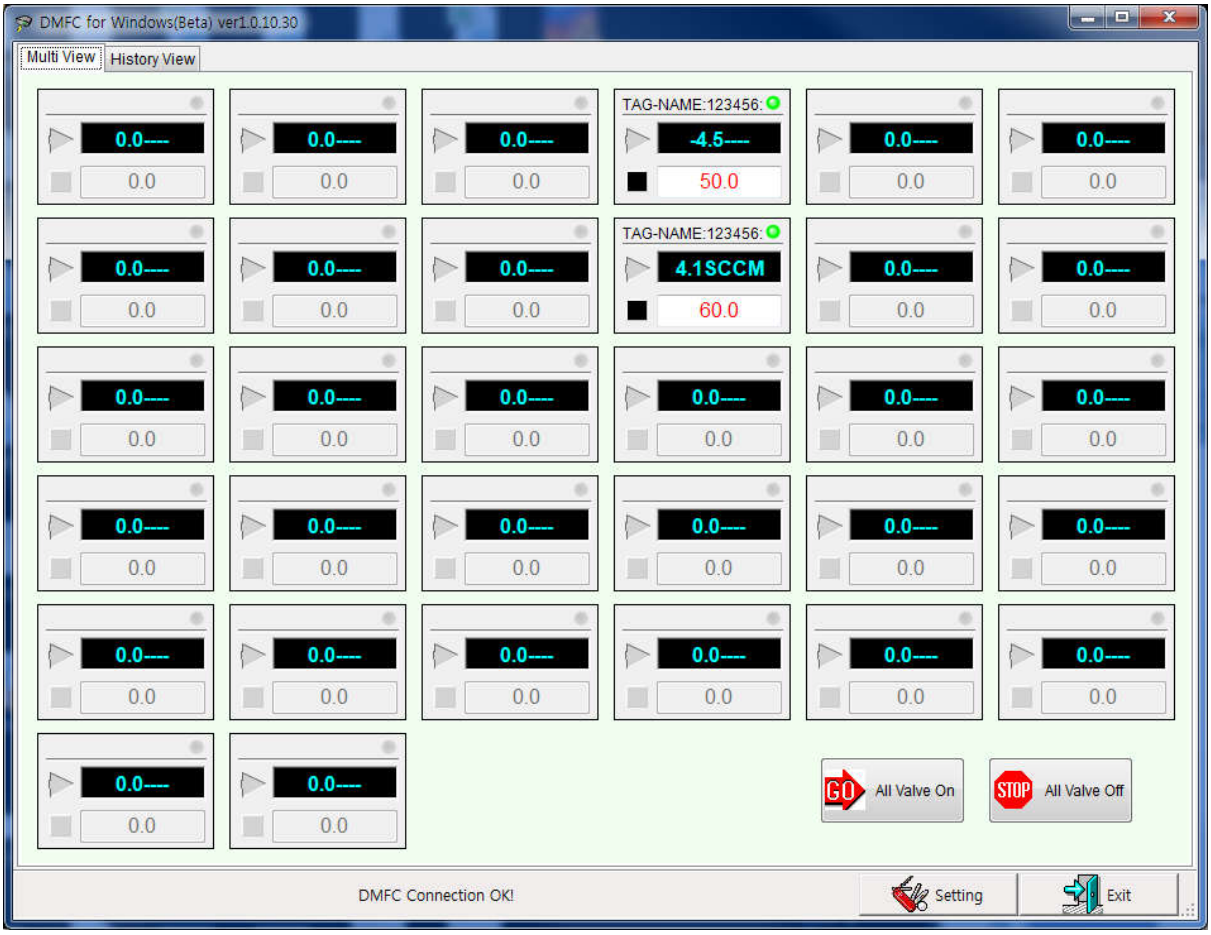
Appendix C. Status bar 12

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DMFC for Windows is

It's a bundle program for the Line Tech DMFC users. It has limited and basic functions for the DMFC/DMFM users. For the more versatile, has a specific function(ex, scheduling), and/or integrated with other device and instruments program, please contact our dealers or us.

Following is a program screen.



<Figure 1. Program running screen >

System requirements

- Pentium compatible 32bit(x86) or 64bit processor
- Windows 7/8(32bit 또는 64bit), Windows XP
  - Windows 8/8.1/10 OS is not yet tested.
- 256MB RAM
- Screen resolution 1024x768
- Microsoft >NET Framework 3.5 (included at distribution package)

Communication hardware

To communicate with DMFC, RS-485 serial communication port is required. If user's PC/Laptop doesn't have this port, use USB to serial(RS-485) converter.

Software functions

- Set up and monitor the flow rate up to 32 DMFC.
- Simultaneously on/off the flow of DMFC up to 32.
- Recent 1minute flow rate chart of an DMFC is available.
- Check the configurations of each DMFC.
- Serial port search and listing, DMFC search and listing.

Parameters Setting

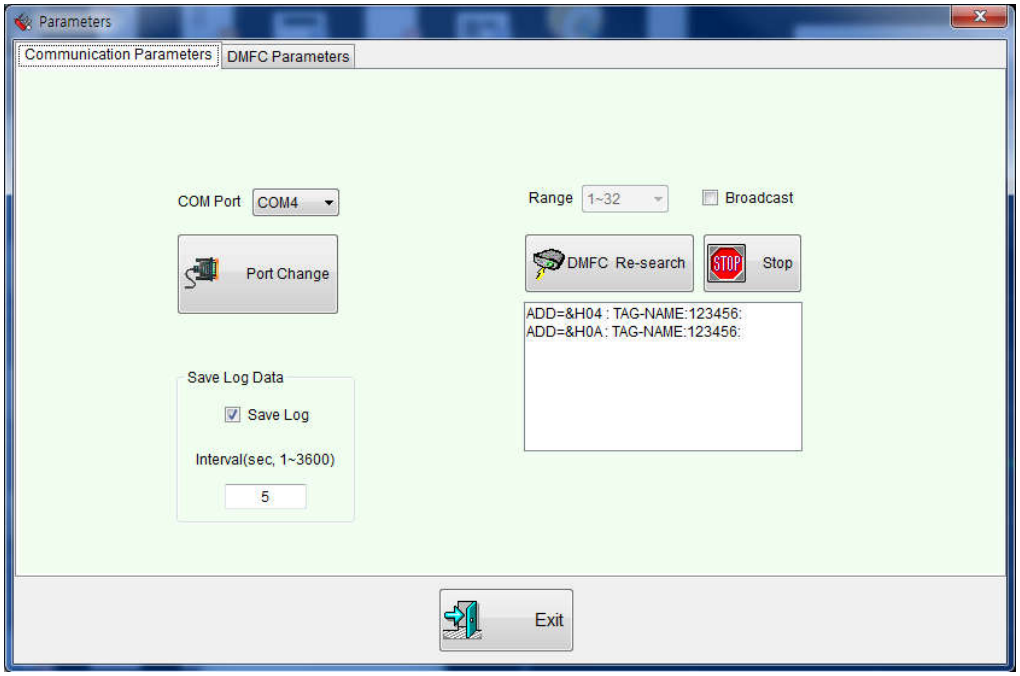
When you first run this program, you must configure the serial port by pressing bottom right [Setting] button.



<Figure 2. Setting button and Exit button.>

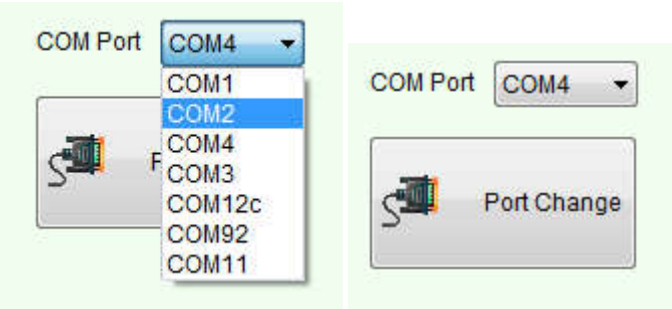
Communication Parameters Tap

At this tap, you can select/change the serial port and search the connected DMFC.



<Figure 3. Com port setting and DMFC searching tap.>

When you run DMFC for Windows, it automatically search the serial ports on your computer. Press the COM Port dropdown menu and select the serial ports, and save current port setting by pressing [Port Change] button.

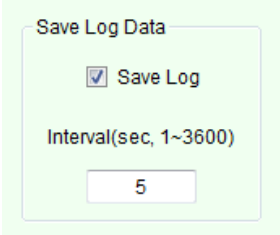


<Figure 4. Serial port selection and change. >

!	This port setting is done only once when you first run this program. Or, when the serial port number is changed.

Data save function

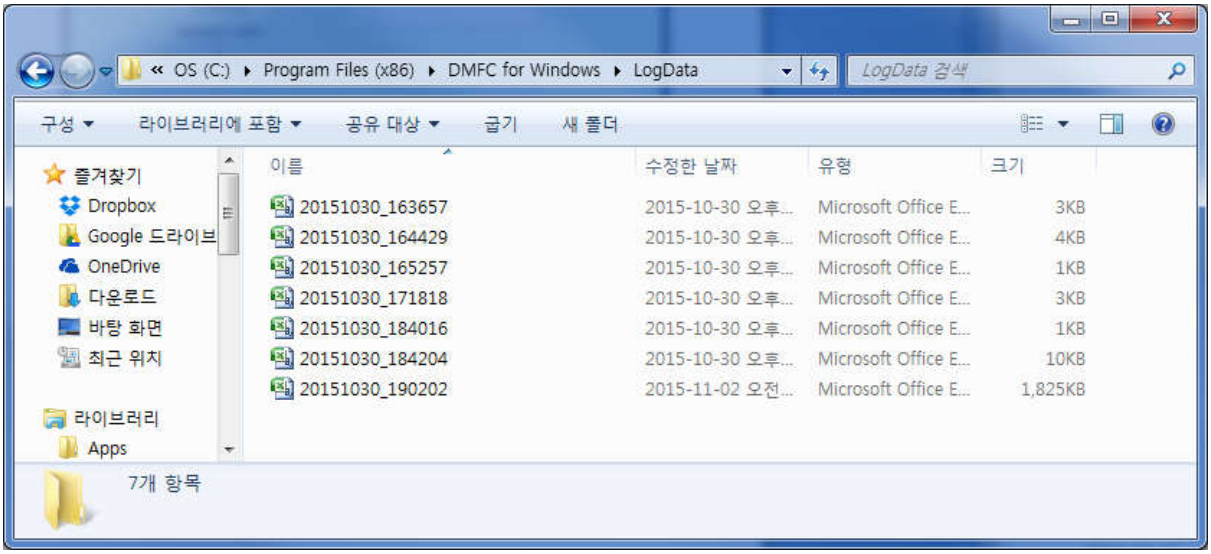
DMFC for Windows program has flow data auto save function.



<Figure 5. Log data save function and set the interval of save(second) >

If the "**Save Log**" check button is checked, the flow data are saved with the user set interval(ex. 5 seconds).

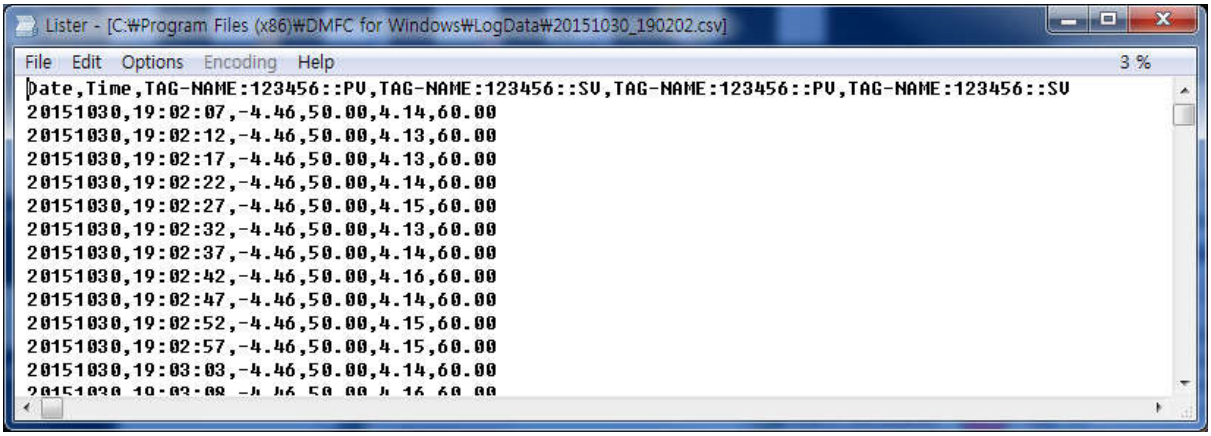
The flow data files are saved to subfolder "LogData" at which DMFC for Windows installed folder(ex. C:₩Program Files₩DFMC for Windows). The file has the "program start time.csv" format.



<Figure 6. Flow data files>

The CSV(comma separated value) file is an text file with each data are separated with a comma. The files in this CSV format easily imported and manipulate with programs such as MS Excel®.

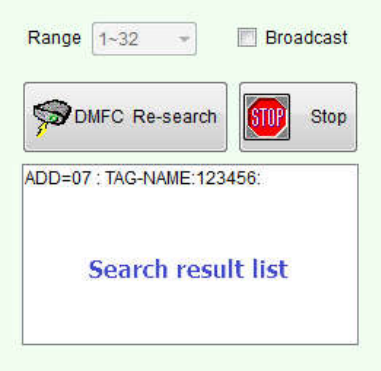
This CSV format file, you can easily check it's contents with generic text file editors.



<Figure 7. CSV file content>

Re-search function

Searching the DMFC connected to the serial port by pressing the [DMFC Re-search] button. DMFC for Windows only searching the address range from 1 to 32 at the specified serial ports, and add the devices found in the searching in turn. While re-searching, user can stop the re-search at any time by pressing the [Stop] button.



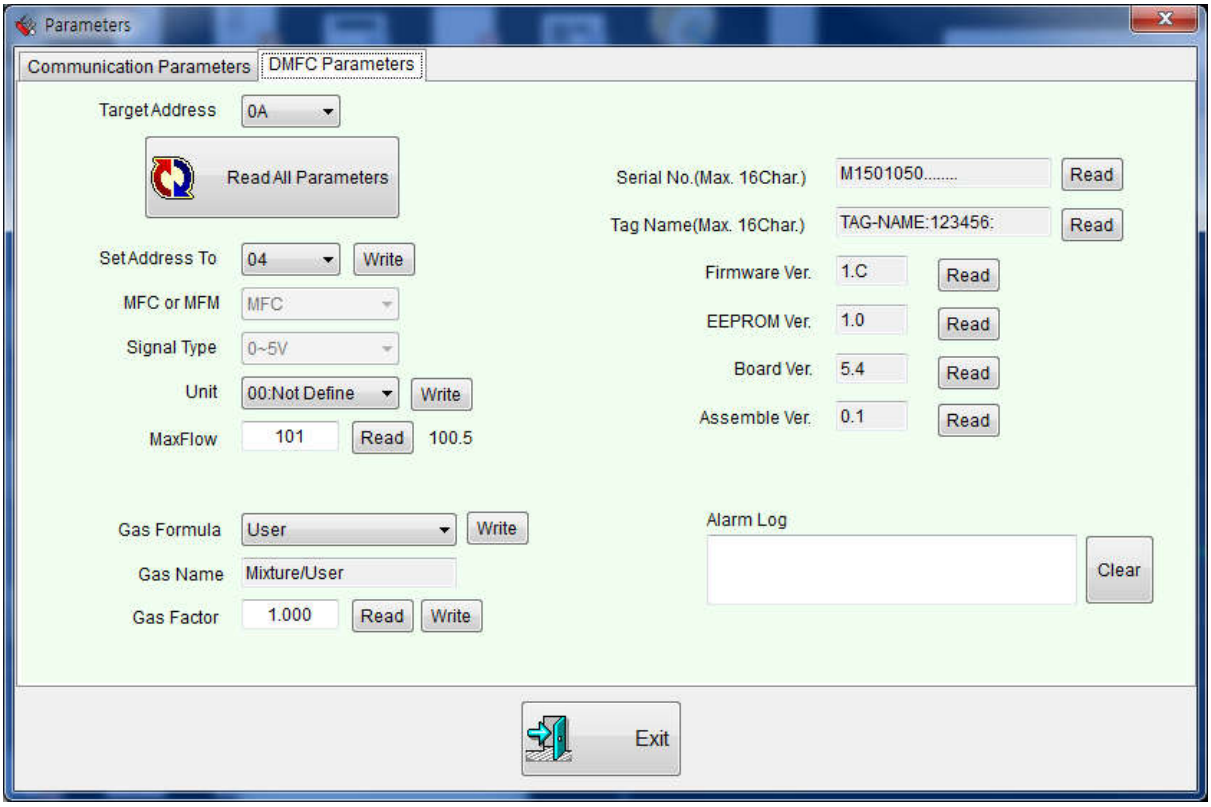
<Figure 8. DMFC search result list>

If you're using only one DMFC/DMFM, check the [Broadcast] check box and press the [DMFC Re-search] button. It reduces the searching time.

!	This re-search is done only once, when you first run this program, when you change the serial port number, or when a DMFC is added or removed from the system.

Parameter configuration tap

At [DMFC Parameters] tap, you can confirm and change the DMFC/DMFM's parameters Some parameters are read only.



<Figure 9. DMFC parameters read and set>

If more than one DMFC/DMFM is connected to RS-485 communication line, select the address of the DMFC/DMFM , and click the [Read all parameters] button.



<Figure 10. DMFC address select and read all parameters>

Other parameters of this tap is .

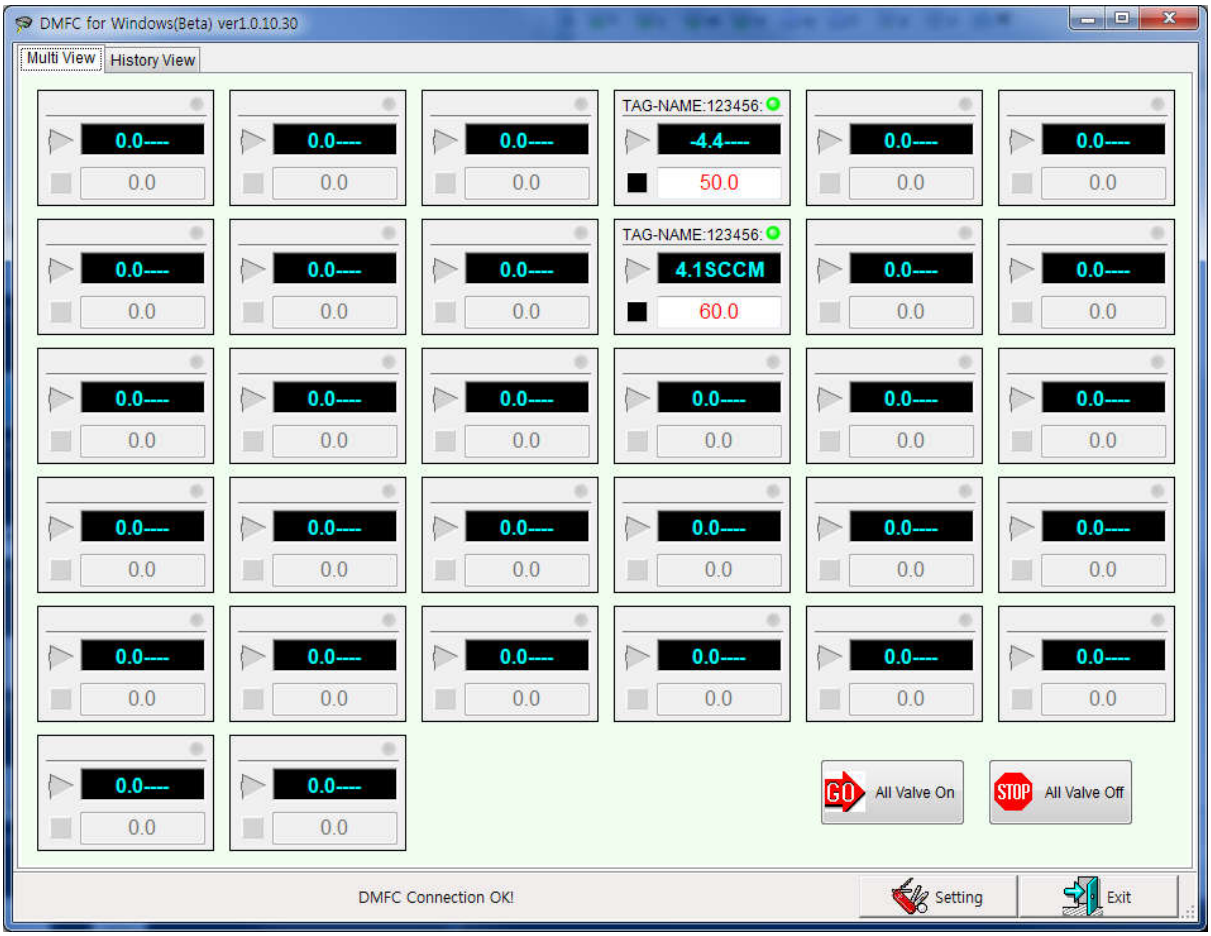
Parameters	R/W	Description
Set Address to	R/W	Shows the address of DMFC and user can change the address. To change the address, select the address and click the [Write] button. The address range of the device at this program is 01~32(decimal).
MFC or MFM	R	Shows the type of the device
Signal Type	R	Shows the analog signal type of the device
Unit	R/W	Shows and change the flow rate unit. To change the unit, select the unit at the drop down menu and click the [Write] button
MaxFlow	R	Shows the maximum flow rate of the DMFC/DMFM
Gas Formula		Shows the gas formula
Gas Name	R	Shows the Gas Name
Gas Factor	R/W	Shows the Gas factor. For the gas mixture, user can edit this value.

Parameters	R/W	Description
Serial No.	R	Shows the serial number of the device at the address
Tag Name	R/W	Shows the tag name of the device at the address
Firmware Ver.	R	Shows the firmware version of the device at the address
EEPROM Ver.	R	Shows the EEPROM format version of the device at the

		address
Board Ver.	R	Shows the PCB's number of the device at the address
Assemble Ver.	R	Shows the production number of the device at the address
Alarm log	R	Lists the errors while communicate with DMFC. To clear, click the [Clear] button.

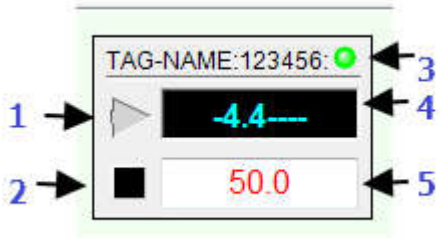
Main window's Multi View tap

At [Multi View] tap, it shows 32 DMFC/DMFM at one window.



<Figure 11. DMFC for Windows program **Multi View** tap>

Each small panel shows following information.

	1. Flow on button 2. Flow off button 3. Displays TAG-NAME 4. Current flow. If click here, change current [Multi view] to specific channel's [History View] 5. Set flow rate.

!	Click current flow, change current [Multi view] to specific channel's [History View], and vice versa. Refer the View Change section.

All Valve On/Off buttons

	1, Set all DMFC to control the gas flow rate. 2. Close all DMFC

History View Tap

Below is the [History View] of DMFC for windows. It displays recent 1minute flow rate plot of specific channel. The red plot is set flow, and the green plot is current actual flow.

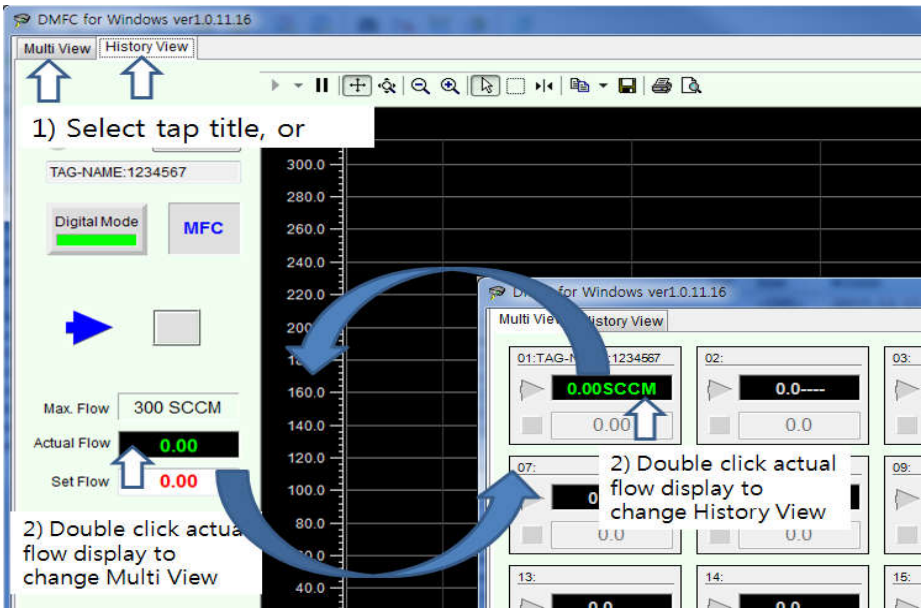


<Figure 12. DFMC for Windows's History View>

	Menu descriptions
(1)	Shows currently selected DMFC's address. The address range is 01~32(decimal number).
(2)	Shows currently selected DMFC's TAG
(3)	Shows currently selected DMFC's mode(digital/analog). Click this button to toggle the mode of DMFC. At digital mode the green led is on.
(4)	Flow on the DM
(5)	Close the DMFC
(6)	Shows currently selected DMFC's maximum flow.
(7)	Shows current flow. Click this panel, the view mode is changed to [Multi view].
(8)	Shows currently selected DMFC's set flow and can change set flow.
(9)	Shows communication errors. The errors are E1, E2. Refer the appendix for the error messages.
(10)	Plot configuration. :: Set the parameters of this flow plot. Only experienced users change the settings please. Refer the appendix for the detailed description for this tool bar.

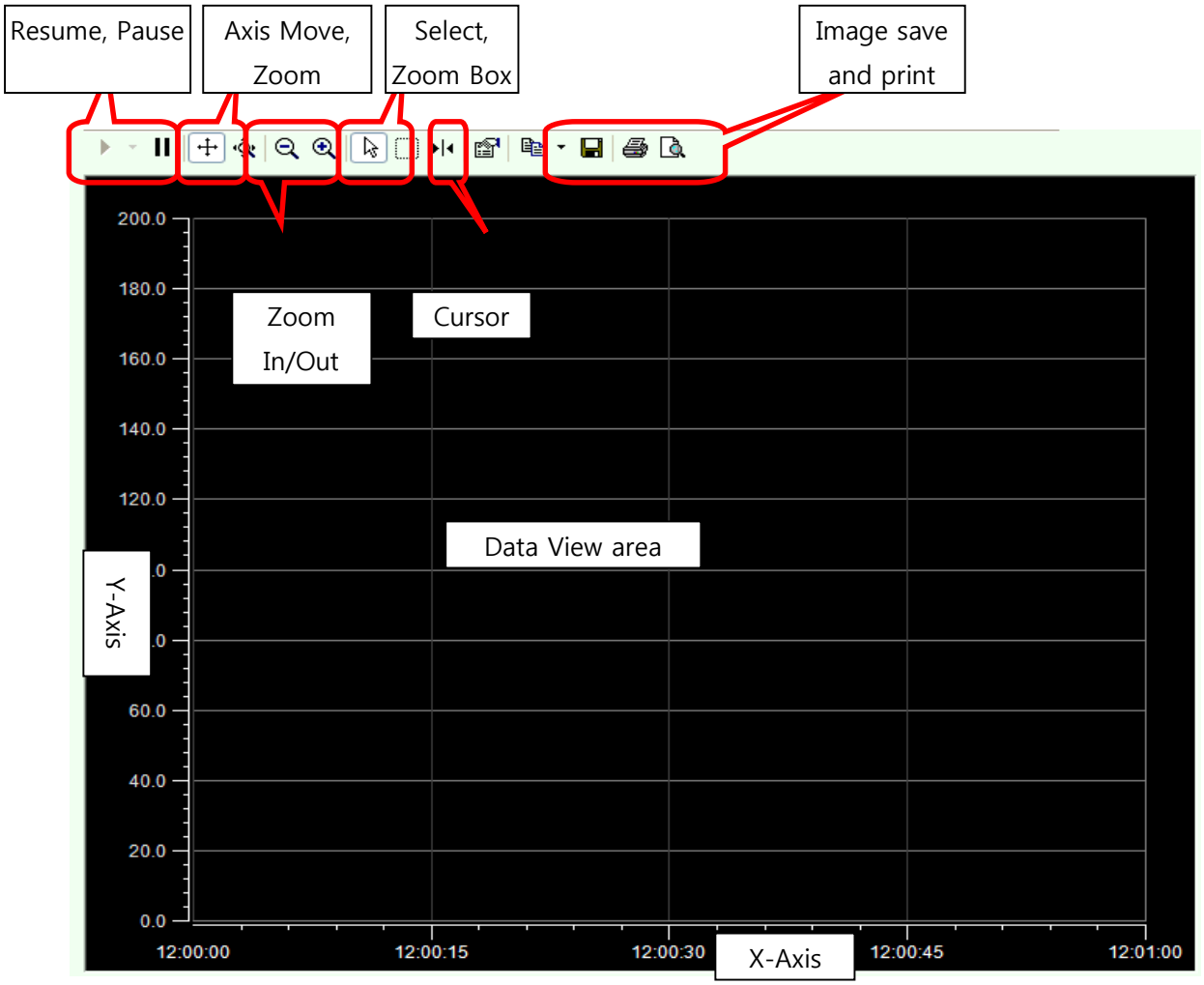
View Change

DMFC for Windows has two view types. To switch another view type, click the tap [Multi View]/[History view], or double click the actual flow display(green number with black background).



Appendix

Appendix A. Plot toolbar function



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- Resume, Pause: Pause and resume horizontal plot scrolling.
- Axis Move, Zoom: Move the X/Y axis and zoom in/out the axis with mouse click and drag
- Zoom In/Out: zoom in or zoom out the plot.
- Select, Zoom Box: Zoom in specific 'Data View area'
- Cursor: Display data value of 'Data View area'
- Copt to Clipboard: Copy current view plot as an image to OS's clipboard.
- Save to File: Save current view plot as an image file.
- Print: Print current view plot
- Preview: Preview of printing current view plot

Appendix B Status and Error number

Status and error

Status	Description
0	Normal
E1	No such a commands
E2	Not authorized command.
E4	Reserved
E8	reserved

Appendix C. Status bar

DMFC for Windows' status bar shows communication error messages.
Following message means there's no response form the DMFC with address 21 for the command 0x04 (read current flow with the flow units).

